

Course 4 · Week 2 — ML done honestly

Cheatsheet — biostats_courses

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Tree-based models

Model	R	Strengths
CART	<code>rpart::rpart</code>	interpretable single tree
Random forest	<code>ranger::ranger</code>	robust, OOB error, variable importance
XGBoost	<code>xgboost::xgb.train</code>	top tabular performance, careful tuning
LightGBM	<code>lightgbm::lgb.train</code>	fast on large data

```
library(ranger)
fit <- ranger(y ~ ., data = df, importance = "permutation",
             num.trees = 500, mtry = floor(sqrt(ncol(df) - 1)))
fit$prediction.error # OOB error
```

Interpretability

```
library(DALEX); library(iml)

ex <- explain(fit, data = df |> select(-y), y = df$y)
ip <- model_parts(ex) # permutation importance
pd <- model_profile(ex, variables = "x1") # partial dependence
sh <- Shapley$new(Predictor$new(fit, data = df, y = df$y),
                 x.interest = df[, 1]) # SHAP
```

PDP assumes feature independence; if features are correlated, use ALE plots.

Tabular neural networks with torch

```
# sketch only — typically in a loop on GPU
library(torch)
nn <- nn_sequential(
  nn_linear(p, 64), nn_relu(),
  nn_linear(64, 1)
)
opt <- optim_adam(nn$parameters)
```

On small biomedical tabular data, boosted trees usually beat NN. Use NN when you need end-to-end training with images, sequences, or text.

tidymodels pipeline

```
library(tidymodels)
rec <- recipe(y ~ ., data = df) |>
  step_normalize(all_numeric_predictors()) |>
```

```

step_dummy(all_nominal_predictors())

mod <- rand_forest(mtry = tune(), trees = 500) |>
  set_engine("ranger") |> set_mode("classification")

wf <- workflow() |> add_recipe(rec) |> add_model(mod)
cv <- vfold_cv(df, v = 5, strata = y)
tuned <- tune_grid(wf, cv, grid = 20)
collect_metrics(tuned)

```

Decision rule for Week 2

- Tabular, < 100k rows → gradient-boosted trees.
- Need explanations → SHAP + PDP, validate with held-out.
- Images / sequences → CNN / transformer, use `torch`.
- Reproducibility → `tidymodels` pipeline + fixed seed + locked recipe.

Common pitfalls

- Reporting feature importance from a model trained on the full data.
- Interpreting PDPs when features are strongly correlated.
- Claiming NN superiority without CV on the same split as a tree model.
- Forgetting to set a seed for any tune / split operation.

Further reading

- Biecek & Burzykowski, *Explanatory Model Analysis*.
- Molnar, *Interpretable Machine Learning*.